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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
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Key Personnel Changes/Previous PI:	Principal Investigator (PI) change from Mark Shelhamer to Michael Schubert in 2013. Continued by "Sensorimotor Assessment and Rehabilitation Apparatus: Procedures and Equipment (PI=Schubert)" with Michael Schubert as PI; same grant number NNX10AO19G and period of performance (8/1/2010-7/31/2014) in NASA Shared Services Center information; for Task Book purposes, period of performance with Schubert will be 8/1/2013-7/31/2014.		
COI Name (Institution):	Schubert, Michael (Johns Hopkins University)		
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Task Description:

Long-duration flight leads to sensorimotor problems which can be critical during landing, rendezvous, and operations on other planetary surfaces. While specific sensorimotor effects have been identified, it is not known which ones have the most adverse impact, or how best to assess them and apply appropriate rehabilitation procedures. NASA's current goal in addressing this situation is to develop a means to assess sensorimotor function rapidly with a portable device, so that an astronaut can make a determination as to whether or not he or she is impaired enough to affect mission safety or success before undertaking a demanding task (piloting, landing, reentry, egress, EVA, tele-operation, etc.). Accordingly, the goal of this project is to develop a portable hand-held device that will allow a single crewmember to assess his/her sensorimotor function in no more than 20 minutes. This is to be accomplished with a judicious choice of which sensorimotor functions to assess, and careful design to obtain the maximum data in the minimum time. The device and procedures being developed to meet these requirements are based on a tablet computer and body-mounted motion-sensor units. Through a set of simple software routines, a rapid assessment of sensorimotor capabilities will be made. The final device will be small, require little power and space, and provide what is essentially a self-contained sensorimotor lab/clinic. We term this device and its embedded software the Sensorimotor Assessment and Rehabilitation Apparatus (SARA).

SARA has been designed to measure the following functional behaviors that are relevant to safe and effective functioning and are also affected by long-duration space flight:

1. Vestibulo-ocular function during pitching head movements (eye movements in response to head movements, a direct measure of vestibular function and canal-otolith integration, and highly relevant for gaze stabilization). We continue to refine our method of subjective nulling of apparent visual motion during head movement (vestibulo-ocular nulling: VON).

2. Vestibulo-ocular function as mediated by the otolith organs. We use a binocular display with lines set by the subject to appear collinear or parallel, to measure vertical skew and disconjugate torsion, which are measures of otolith asymmetry.

3. Postural control, by measuring body sway with eyes closed. This test is enhanced by having the subject make pitching head movements.

4. Locomotion, by measuring alterations in the coupling between pitching head movements and vertical body motions during walking.

NOTE (Ed., 6/13/2018): Principal Investigator (PI) change from Mark Shelhamer to Michael Schubert in 2013. Continued by "Sensorimotor Assessment and Rehabilitation Apparatus: Procedures and Equipment (PI=Schubert)" with Michael Schubert as PI; same grant number NNX10AO19G and period of performance (8/1/2010-7/31/2014) in NASA Shared Services Center information; for Task Book purposes, period of performance with Schubert will be 8/1/2013-7/31/2014.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our sensorimotor assessment apparatus has been implemented on a tablet computer and motion sensors with Bluetooth wireless link. This provides real-time data acquisition and processing of three linear accelerometers and three rate sensors from each of three sensors. This meets the need for rapid non-invasive screening of vestibular and sensorimotor function in settings where special expertise and equipment is not available. Applications include assessment of passengers before and after parabolic and sub-orbital flight, assessment/rehabilitation of vestibular patients away from specialized centers, and assessment of crew immediately on return from ISS (on R+0 in Russia).
Possible uses for SARA:

• Go / No-Go decision before demanding task:

o After landing (moon, Mars, NEO), o Before EVA (deep space);

• Data on acute deficits upon return from long-duration stay on ISS:

o Inform countermeasure priorities;

• On board ISS:

o Before EVA, o Correlate assessment with performance;

• Field use: rapid evaluation with minimal resources:

o Centrifuge studies, o Before/after suborbital flight, o Immediately upon return from ISS – to evaluate sensorimotor disturbances and inform countermeasure development (R+0 testing);

• Lab adaptation studies:

o Multi-system approach, o Determination of individual "sensorimotor signature," o How do systems interact during perturbations and adaptation.

We continue to develop software based on our adopted hardware system: tablet computer with Android operating system programmed in Java, and motion sensors from Shimmer Research (self-contained linear and angular motion sensors with Bluetooth link).
The assessments implemented to date include:

1. Vestibulo-ocular nulling (VON). A head-mounted motion sensor controls, through a variable gain set by the subject, the motion of a dot on the display. The subject sets the gain so that, during pitching head movements, the dot appears stationary in space. This means that it is stationary on the retina, and that dot motion matches eye motion in space. In this way, any deficiency in the VOR can be determined (i.e., how much the eyes move when the head moves).

2. Vertical and torsional skew. Red and blue lines are presented on the tablet display, and the subject wears red/blue spectacles, so that each eye sees one line. The subject is instructed to set the lines so that they appear either adjacent (in

Task Progress:

the case of vertical offsets) or parallel (for torsional offsets). This provides a subjective measure of vertical and torsional skew.

3. Posture. The subject performs the VON test while standing. This provides an additional test of vestibulo-ocular function in a natural condition, while the head movements provide a perturbation to the postural system. Head and trunk sway are measured with motion sensors.

4. Locomotion. The subject walks a predetermined path, with and without head movements (yaw or pitch), while motion sensors detect head and trunk motion.

Recent progress on implementation has been greatly aided by the hiring of a student programmer proficient in Java, and two undergraduates. The most significant changes so far have been in skew assessment. Line colors on the tablet screen have been changed to more closely match those of the red/blue goggles, providing better binocular separation of the stimulus (one line seen by each eye). The lines have also been placed adjacent to each other, so that when they are properly aligned they are collinear rather than overlaid. This avoids the problem of one line disappearing under the other when they are aligned. Progress has also been made on standardizing viewing distance from the eyes to the tablet screen, which is difficult in a setting such as parabolic flight.

Relatively less work has been done on the VON procedures, although we have given this some thought as we continue to use this methodology in our lab studies. After finalizing our approach to skew, efforts will be dedicated to VON for VOR assessment. Another implementation issue that remains to be addressed is synchronization of data streams across different motion sensors. Since each sensor unit has its own clock, the data streams from multiple sensors can lose time synchrony. We are exploring software solutions to this problem.

Future tests to be considered for implementation include:

1. Manual control. Eye-hand tracking, following a moving point on the display with a joystick controller.

2. Vertical writing, an otolith assessment based on latent righting reflex.

3. Saccade targeting, or alternative assessment of spatial reference frame.

A patent application has been filed for the vestibulo-ocular and skew tests (application No. 13/229,403, Apparatus and Method for Assessing Vestibulo-Ocular Function).

Laboratory testing

VOR adaptation & VON validation (adopted from 2013 HRP poster by K Beaton)

One of the key scientific motivators behind SARA development is the ability to make rapid assessment of multiple sensorimotor systems with minimal time and equipment. In this vein, we have been performing laboratory experiments that make use of some of the SARA capabilities, that indicate improvements to their implementation, and that provide a scientific basis for the selection and development of appropriate algorithms for the analysis of the resulting multiple data streams. Here, we considered interactions between the vestibulo-ocular and postural systems during pitch-plane adaptation to a visual-vestibular disturbance. We also evaluated the extent to which adaptation transfers from the pitch plane to the yaw plane.

Normal healthy subjects wore 0.5X minifying lenses for 20 minutes, during which they performed pitch-plane gaze-stability exercises to adapt their VOR gain. During adaptation, subjects were exposed to one of two postural stances: (1) SIT – seated on an upright chair with feet flat on the floor, or (2) STAND – standing on 3-inch-thick foam with heels and toes together and arms crossed. VOR adaptation was assessed through simultaneous eye and head movement recordings (EyeSeeCam, Munich Germany), as subjects made active pitch and yaw head rotations while imagining a distant visual target in the dark. VOR gain was defined as the ratio of peak-to-peak eye position to peak-to-peak head position. During adaptation, VOR gain was measured every five minutes in four conditions: (1) pitch plane SIT, (2) yaw plane SIT, (3) pitch plane STAND, (4) yaw plane STAND.

All subjects adapted to the minifying lenses, as verified by reduced VOR gain following adaptation. In the pitch-plane tests, subjects displayed the largest amount of adaptation when tested in the condition most similar to that used during adaptation: those who adapted in SIT showed the most adaptation during SIT tests, and those who adapted in STAND showed the most adaptation during STAND tests. This represents a type of contextual adaptation. On the other hand, there was a small amount of transfer of adaptation from the pitch plane to the yaw plane in most subjects, indicating generalization.

Variability in VOR gain was larger during pitch than yaw, perhaps because the sensorimotor system is accustomed to compensating for pitching head movements during locomotion and so allows for some amount of normal deviation in that direction. Results suggest that the rate of VOR adaptation is faster during STAND verses SIT adaptation; this may be because the central nervous system as a whole must compensate for the more challenging balance condition quickly to minimize fall risk.

We hypothesize that subjects experienced two forms of adaptation during this experiment: context-specific adaptation, in which postural stance served as the context cue, and adaptive generalization, in which subjects were able to transfer some of the adaptation from the pitch plane to the yaw plane. How the nervous system decides which responses to generalize and which to make specific is an important question that we are now pursuing. The answer has important implications for the adaptation of multiple sensorimotor responses in different gravity environments.

Predicting adaptation ability

In the course of the preceding VOR-adaptation experiment, we pursued an intriguing finding first made in our parallel investigations on saccade adaptation (under NIH and NSF support). In the saccade study, we found that there are strong statistical correlations between the amplitudes of consecutive predictive saccades; these saccades are made to two periodically paced visual targets, and occur with low latencies so that each saccade must be programmed in anticipation of the next target occurrence. These inter-trial correlations decay as a power-law rather than exponentially, which is assessed by taking the power spectrum of the sequence of primary-saccade amplitudes and finding the slope of the spectrum. That the correlations decay as a power law indicates that the amplitudes form a fractal time series.

Each subject in our saccade experiment generated a sequence of 300 such predictive saccades, and the spectral slope

(beta) was found for each subject, as a measure of the strength and decay-rate of the inter-trial correlations. In the same session, each subject was then presented with a conventional double-step adaptation paradigm: a target is presented which moves as the eyes approach it. Since vision is essentially masked during a saccade, the subject perceives not that the target has moved but rather that the saccade has been inaccurate. After a series of such trials, saccades are made with amplitudes appropriate for the displaced target rather than for the initial target location. The key finding is that the fractal structure of the predictive saccades is strongly correlated with the rate of adaptation. In other words, stronger correlations within a sequence of predictive saccades is associated with faster adaptation. We believe that this is because the inter-trial correlations reflect the ability to effectively store and process information on previous saccade errors, and to use that information in generating subsequent movements. This surprising finding suggests that we may be able to forecast a person's ability to adapt in a particular sensorimotor task.

To extend this result to an area more relevant to some of the vestibular concerns of space flight, we applied a similar analysis to data from the VOR-adaptation experiment described above. At the beginning of each experiment session, the subject performed a series of approximately 400 rapid steps of head position (turning the head in yaw by approximately 30 deg each time), paced at 1 Hz. VOR gain was found for each of these trials (each head movement), as eye velocity divided by head velocity at the time of peak head velocity. This sequence of VOR gains was examined for inter-trial correlations, as above for saccades, by finding the slope (beta) of the power spectrum of the sequence of VOR gains.

After this initial set of head steps, VOR gain-decrease adaptation was carried out as above with minifying lenses. The extent of adaptation was determined as the difference between VOR gains before adaptation and after 20 minutes of adaptation. As for saccades, there is a strong relationship between the structure of the correlations in inter-trial VOR gains (as measured by beta), and the extent of adaptation.

This strong correlation is promising, and suggests that we may be able to predict the ability to adapt to vestibular disturbances in a given subject. There is, however, an important difference between this VOR result and that from saccades: the correlations are in opposite directions. In other words, here for the VOR, the stronger the inter-trial correlations (larger beta) the worse the adaptation. Further experiments and computational refinements are needed to understand this, and we are pursuing those now.

We believe we can explain this particular relationship as follows. For saccades, amplitude error of a given movement is available only after the saccade ends, and it can only be used to affect the programming of the following saccade (not the current one, just ended). Thus there is a strong need to store saccade error information across trials, leading to strong correlations. In the VOR, on the other hand, error information is available constantly and in real time, through slip of images on the retina if the VOR is improperly calibrated. Thus, for the VOR, the best way to get reliable current information on the state of the system (stimulus-response relationship) is to not store any information from the past but rather to pay attention only to what is happening at the moment. The more effectively the system can do this, the weaker are the correlations between trials, and the better the adaptation.

We believe that this finding can be of fundamental importance in determining adaptation ability in a given sensorimotor task, and in tailoring training and adaptation procedures accordingly for each person.

Parabolic flight testing

General observations (adopted from 2013 Shelhamer HRP presentation)

VON: The subject's task is simple and easily accomplished, although screen glare is a distraction. The nulling task overall is not as intuitive as we would like: the subject must ignore the frame of the tablet computer in judging target motion. We again verified that pitch VOR gain is higher in the high-g phase of flight, although scatter is large.

Skew: The subject's task was rapid, easy, and intuitive. Torsional measures showed a small disconjugacy. Vertical skew measures were initially inconsistent and showed no systematic effects, but this has been improved (see below).

Posture: Motion sensors provided data of sufficient quality for postural assessment. Posture is stable in baseline testing: there is very little body motion during head movements. Upon early exposure to lunar g (non-adapted), there is considerable body motion in all three axes during head motion, especially in yaw as the subject attempts to maintain center-of-pressure over the feet in the face of unreliable vestibular information on head/body motion.

Locomotion: Data quality was sufficient to assess locomotion. The wireless link from sensors to tablet computer did not drop out as the subject moved. Locomotion in lunar g is not dramatically altered and does not change much over four flights, although head yaw is more tightly controlled after several flights. The head and body turn together early in flight; after adaptation to lunar g, head motion leads body motion. The latter is more natural, where the head turns first to see where to go, followed by the body turning. The coupled head-body motion seen early resembles that seen in vestibular patients, and again shows that our technology can detect even subtle sensorimotor alterations.

Skew assessment and validation

During flights in April 2013, SARA technology was tested in different g levels, testing improvements in the stimulus as described above. During baseline testing on the ground, roll tilts that activate the otolith organs might be expected to produce skew if there is an otolith asymmetry. In an initial set of subjects there are indeed significant differences between the three head orientations, and although the differences are small the variability is well-controlled, enabling reliable statistical determinations. Testing in the three g levels of parabolic flight (0g, 1g, ~2g) also yields clear differences between conditions.

Different subjects show different patterns of skew and disconjugacy upon head tilt and in altered g. The modeling to explore this, and for relating the baseline results to flight results, is in process.

NOTE (Ed., 5/5/2014): Continued by "Sensorimotor Assessment and Rehabilitation Apparatus: Procedures and Equipment (PI=Schubert)" with Michael Schubert as PI; same grant number NNX10AO19G and period of performance.

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